

SIMETRIJE V FIZIKI

Spontani zlom simetrije

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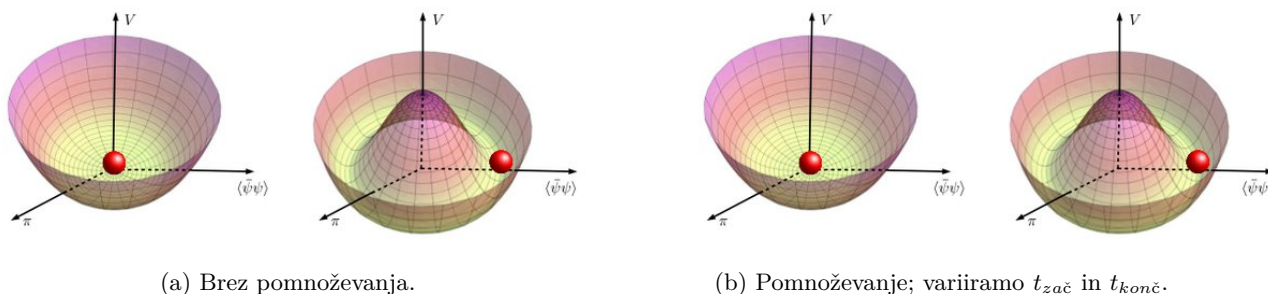
UNIVERZA V LJUBLJANI
FAKULTETA ZA MATEMATIKO IN FIZIKO

1 Globalna simetrija

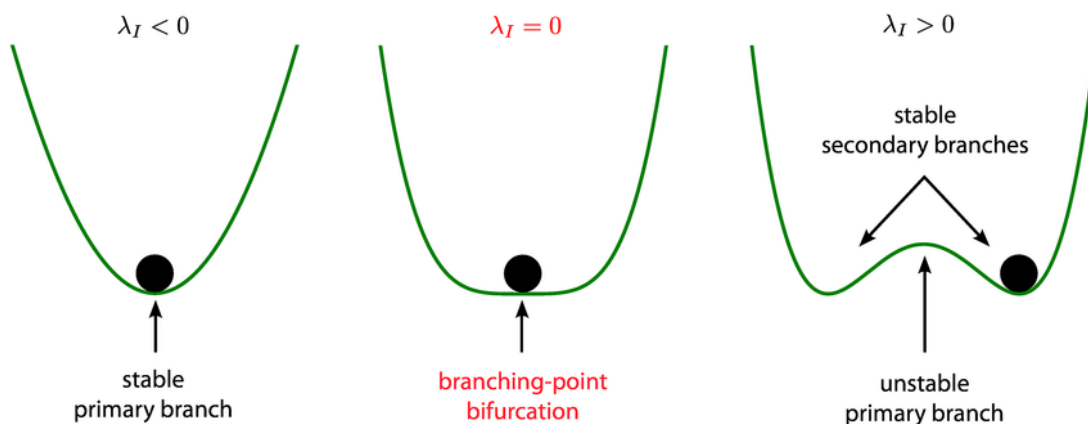
Spontani zlom simetrije, Nambu-Goldstone-ov izrek. Literatura: Gauge theory of elementary particle physics, T-P. Cheng and L-F. Li (pages 144-145).

Spontani zlom $SU(2)_L \times SU(2)_R$ kiralna simetrija, sigma model.

Pioni kot Nambu-Goldstone-ov bozoni, mase nukleonov. mass of nucleon. Literatura: Gauge theory of elementary particle physics, T-P. Cheng and L-F. Li, 144-150;



Slika 1: Napoved elastičnih konstant iz eksperimentalnih meritev.



Slika 2: Abc.

Literatura

- [1] F. Halzen in A. D. Martin *Quarks and Leptons: An Introductory Course in Modern Particle Physics*. Wiley, 1984
- [2] Michael E. Peskin *The T_{EX} Book*, Addison-Wesley Professional. An Introduction to quantum field theory. CRC press, 2018.
- [3] Why do we expect a Higgs boson? Part I: Electroweak Symmetry Breaking *The T_{EX} Book*, Addison-Wesley Professional. url: <http://www.quantumdiaries.org/2011/11/21/why-do-we-expect-a-higgs-boson-part-i-electroweak-symmetry-breaking/>.
- [4] Leslie Lamport (1994) *L^AT_EX: a document preparation system*, Addison Wesley, Massachusetts, 2nd ed.